To make it same, their equivalent fractions are to be found.

Finding the equivalent fractions of $\frac{2}{5}$ and $\frac{1}{3}$ having common denominators are the same as finding the least common multiple of the denominators of the given fractions.

Hence $\frac{2}{5}=\frac{2\times3}{5\times3}=\frac{6}{15}$ and $\frac{1}{3}=\frac{1\times5}{3\times5}=\frac{5}{15}$ So, $\frac{6}{15} > \frac{5}{15}$

Therefore, we can conclude that Madhu has eaten more chocolates.

Note

The process of finding the like fractions of the given unlike fractions can be made easier by finding the common multiples of the denominators of the unlike fractions.

Example 3 Vinotha, Mugilarasi, Senthamizh were participating in the water filling competition. Each one was given a bottle of equal volume to fill water in it within 30 seconds. If Vinotha filled $\frac{1}{2}$ portion of her bottle, Senthamizh filled $\frac{3}{4}$ portion of her bottle and Mugilarasi filled $\frac{1}{4}$ portion of her bottle, then who would get the first, second and third prize?

Solution The equivalent fractions need to be written until the denominator becomes 4 which is the LCM of 2 and 4.

Equivalent fraction of $\frac{1}{2}$ is $\frac{2}{4}$

Vinotha's portion	Mugilarasi's portion	Senthamizh's portion
$\frac{1}{2}=\frac{2}{4}$	$\frac{1}{4}$	$\frac{3}{4}$

Here $\frac{1}{4}<\frac{2}{4}<\frac{3}{4}$. Therefore, Senthamizh would get the first prize, Vinotha would get the second prize and Mugilarasi would get the third prize.

Example 4 Arrange $\frac{2}{3},\frac{1}{6},\frac{4}{9}$ in ascending order.

Solution Equivalent fractions of $\frac{2}{3}$ are $\frac{4}{6},\frac{6}{9},\frac{8}{12},\frac{10}{15},\frac{12}{18},\ldots$

Equivalent fractions of $\frac{1}{6}$ are $\frac{2}{12},\frac{3}{18},\ldots$

Equivalent fraction of $\frac{4}{9}$ is $\frac{8}{18},\ldots$

Therefore $\frac{3}{18}<\frac{8}{18}<\frac{12}{18}$

The ascending order of given fractions is $\frac{1}{6},\frac{4}{9},\frac{2}{3}$.

6th_ Maths _Chapter 1.indd 6 22-10-2019 18:11:04